

ON THE EXPANSION OF $(1 - z_p)^n$

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Friday Seminar
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Why do we care?

Why do we care?

- $(1 - z_p)^n \longrightarrow$ The p -adic evaluation problem.
- Lots of applications in Abstract Algebra
(MATH-450)
and Number Theory.
- Proof technique requires inspection and
induction.

A closer look at $(1-z_p)^n$

$$(1 - z_p)^n$$

→ We control
this
parameter

a p^{th} root
of unity

→ A fixed
prime number.

What is a p^{th} root of unity?

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Algebraic: the solutions to the equation

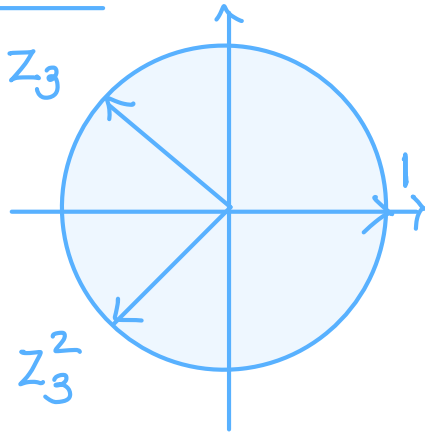
$$x^p = 1, \quad p \text{ is a prime number.}$$

What is a p^{th} root of unity?

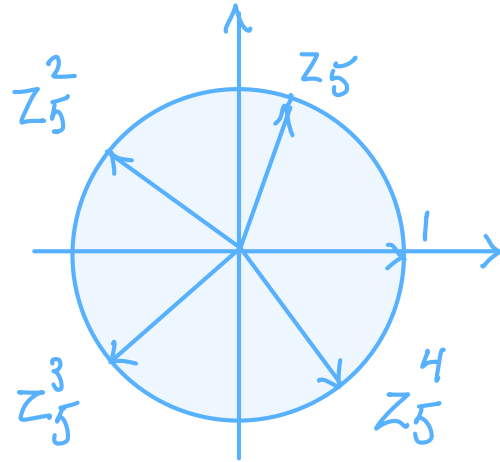
Algebraic: the solutions to the equation

$$x^p = 1, \quad p \text{ is a prime number.}$$

Geometric



$$p = 3$$



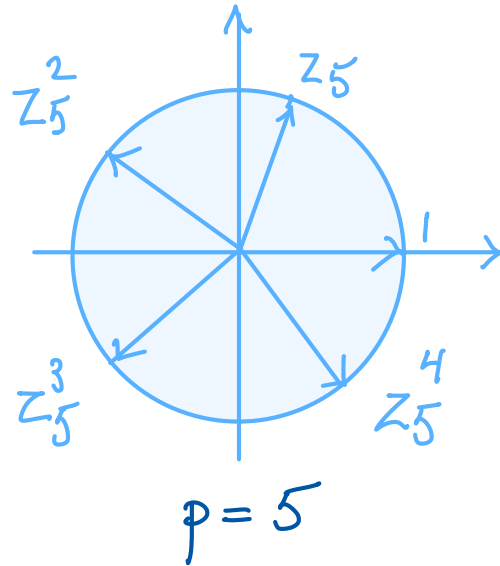
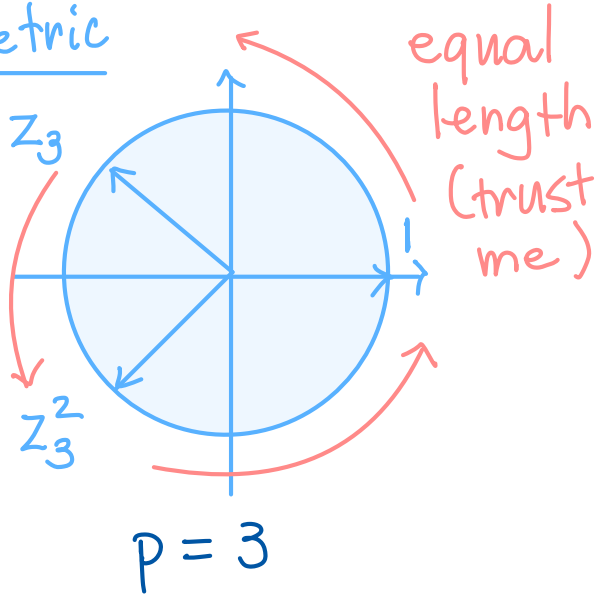
$$p = 5$$

What is a p^{th} root of unity?

Algebraic: the solutions to the equation

$$x^p = 1, \quad p \text{ is a prime number.}$$

Geometric



So, why is z_p^k special...?

$$z_p^k = e^{\frac{2\pi k}{p} i}, \quad i = \sqrt{-1}$$
$$k = 1, 2, 3, \dots, p.$$

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①

Meaning: If we can simplify $(1 - z_p)^n$ in terms of z_p^k , we can compute the expression directly!

Claim:

$$1 + z_3 + z_3^2 = 0$$

$$1 + z_5 + z_5^2 + z_5^3 + z_5^4 = 0$$

$\vdots$

$$1 + z_p + z_p^2 + \dots + z_p^{p-1} = 0$$

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Why? Proof left as an exercise for readers.

Example $(1-z_p)^n$ when $p=3, n=2,3,\dots$

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$$= z_3.$$

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$$(1-z_3)^3$$

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$$= -3z_3 - 3(-1-z_3) \quad \text{by } \textcircled{2}$$

$$= -3 - 6z_3.$$

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In general, for $p=3$:

$$\begin{aligned}(1-z_3)^n &= (1-z_3)^{n-1} (1-z_3) \\ &= c_{0,n} + c_{1,n} z_3\end{aligned}$$

The reduced problem

Instead of looking for $(1-z_p)^n$, we look for the representation coefficients (ROU coefficients).

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How many coef's do we need? $(p-1)$.

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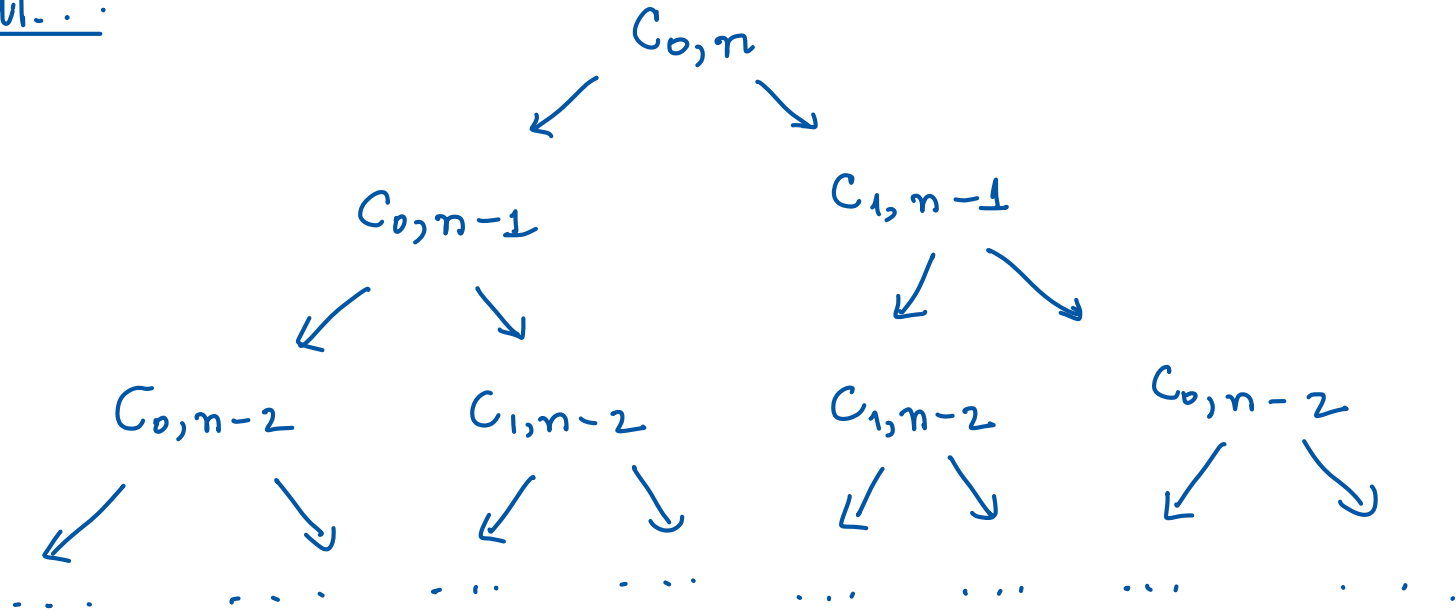
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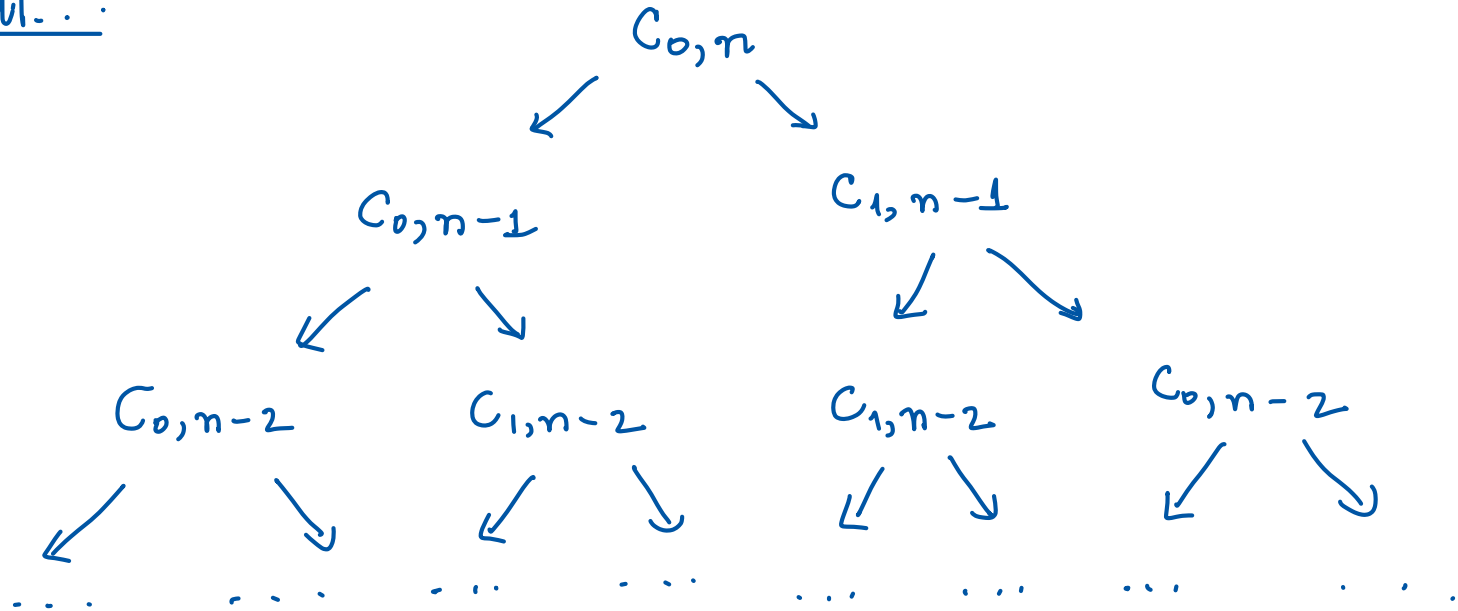
$$\text{For } p=3, \quad c_{0,n} = c_{0,n-1} + c_{1,n-1}$$

$$c_{1,n} = 2c_{1,n-1} - c_{0,n-1}$$

But...



But...



This is exponential arithmetic complexity.

A very insufficient overview of complexity analysis

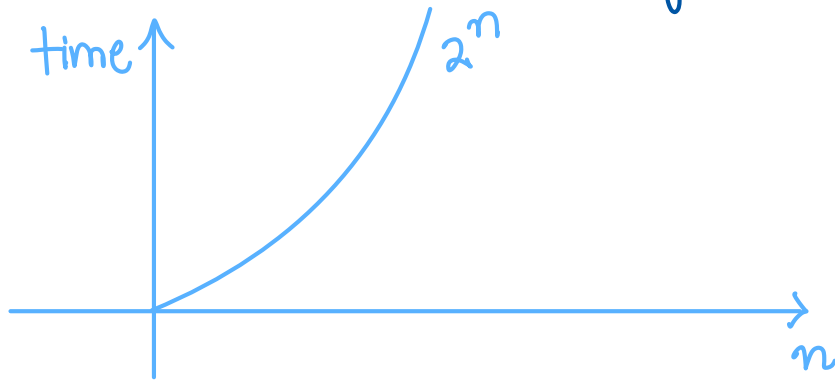
(CSC-371)

A very insufficient overview of complexity analysis

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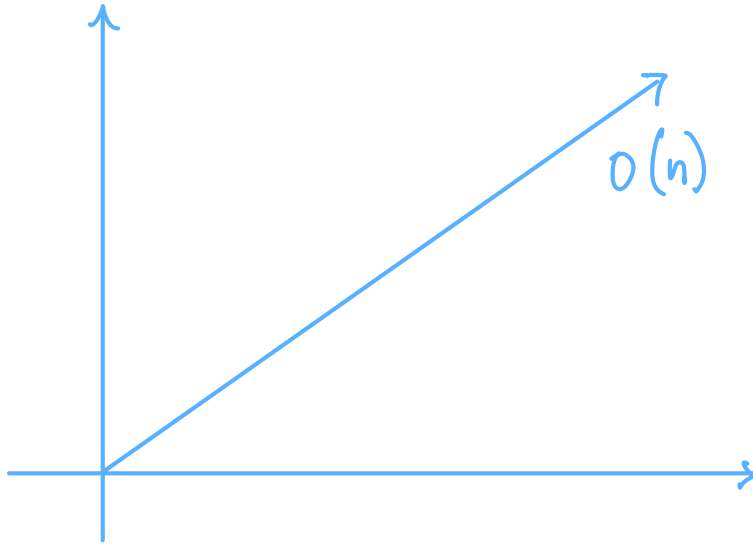
$O(2^n)$

- Exponential Time : If the cost of 1 operation is 2 seconds, then it would take around $2^2 = 4$ seconds to get the final result.

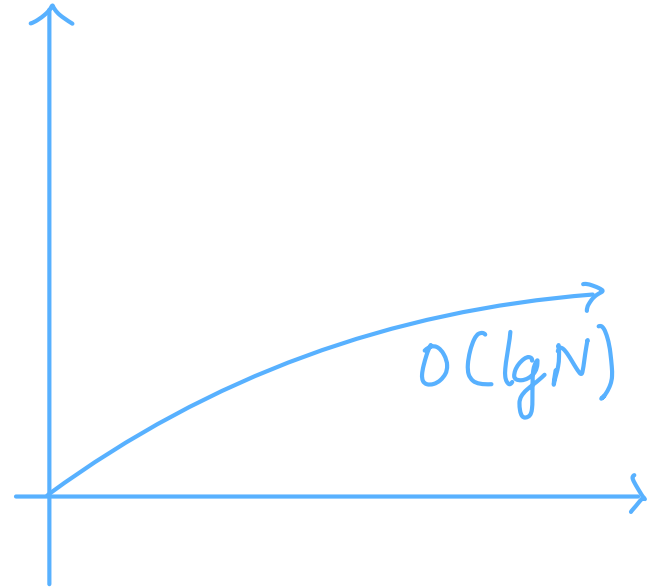


(Our universe is
 $\approx 2^{34}$ years old)

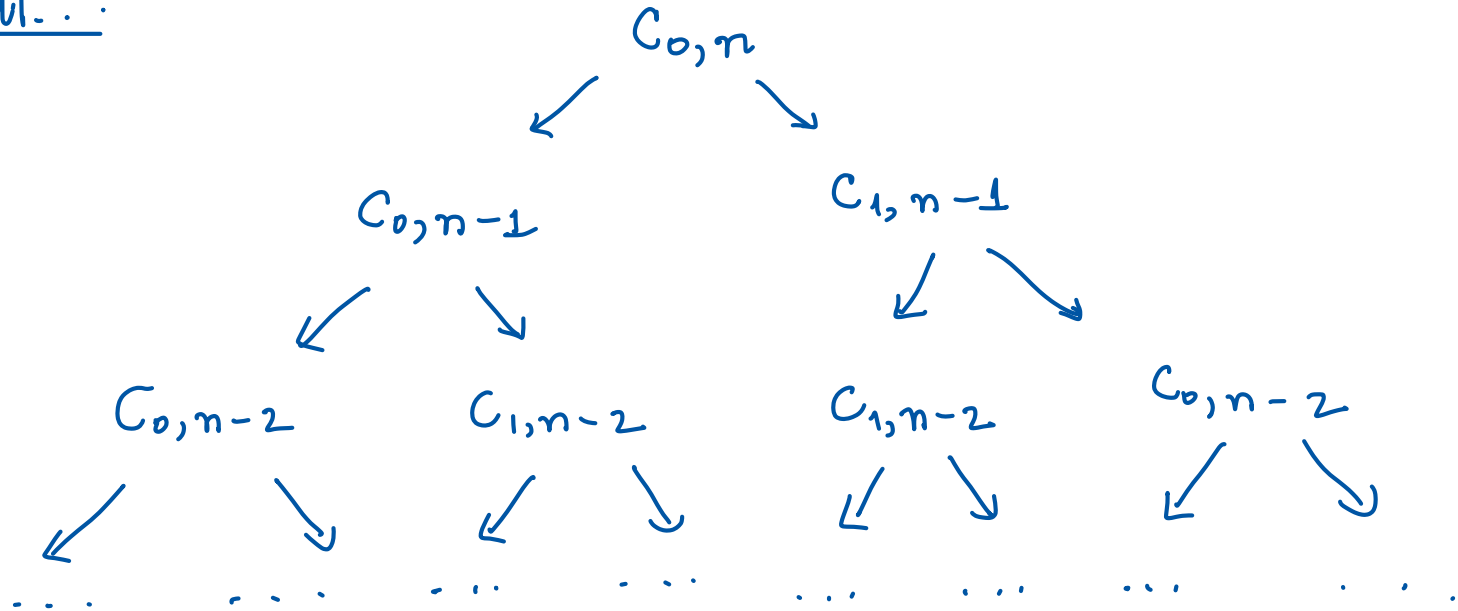
- Linear time $O(n)$



- log time $O(\lg n)$



But...



This is exponential arithmetic complexity.

Goal:

- Computationally: Naive recursion is $O(2^n)$ arithmetic complexity.

Solution: Smart recursion, memoization, dynamic programming (CSC-371).

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Solution: Smart recursion, memoization, dynamic programming (CSC-371).

- Mathematically: Closed form to find the ROU coefficients.

Binomial Theorem

For $a, b \in \mathbb{C}$, $n \in \mathbb{Z}^+ + \{0\}$:

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k} \quad (3)$$

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what is this notation?

The Binomial Coefficients

1

1 1

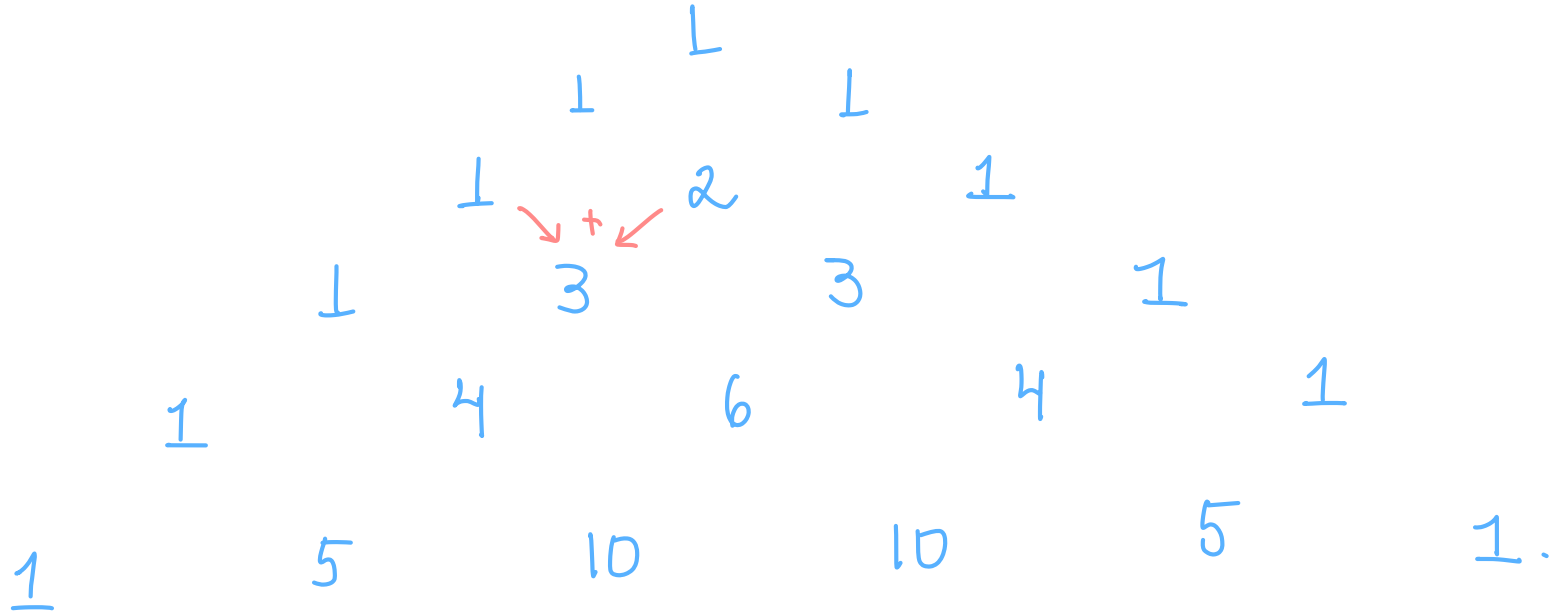
1 2 1

1 3 3 1

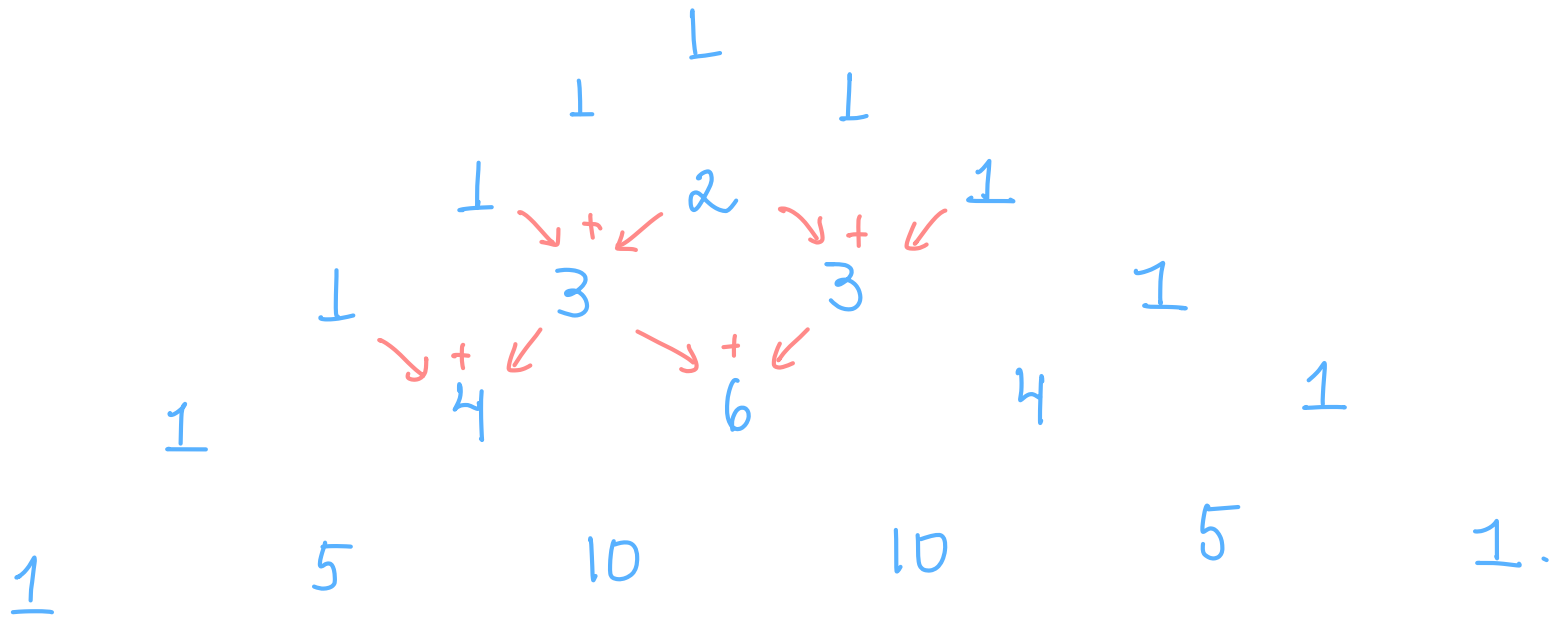
1 4 6 4 1

1 5 10 10 5 1

The Binomial Coefficients



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The Binomial Coefficients

$$\binom{n}{k} = \begin{cases} \frac{n!}{k!(n-k)!} & , 0 \leq k \leq n \\ 0 & , k > n \end{cases} \quad (3)$$

where $0! = 1$

$$1! = 1$$

$$n! = n(n-1)(n-2) \dots 3 \cdot 2 \cdot 1.$$

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Let $n=5$, $a=-z_3$, $b=1$.

$$(1-z_3)^5 =$$

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Let $n=5$, $a=-z_3$, $b=1$.

$$(1-z_3)^5 = \sum_{k=0}^5 \binom{n}{k} (-1)^k z_3^k$$

Binomial Theorem

$$\begin{aligned}(1 - z_3)^5 &= \sum_{k=0}^5 \binom{5}{k} (-1)^k z_3^k \\&= \left[\binom{5}{0} (-1)^0 + \binom{5}{3} (-1)^3 \right] \cdot 1 \\&\quad + \left[\binom{5}{1} (-1)^1 + \binom{5}{4} (-1)^4 \right] z_3 \\&\quad + \left[\binom{5}{2} (-1)^2 + \binom{5}{5} (-1)^5 \right] z_3^2.\end{aligned}$$

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then
collapse
to 1, z_3

Binomial Expansion

For $n \geq 0$, the coefficients $c_{j,n}$ can be calculate as:

$$c_{j,n} = \sum_{k=0}^{\lfloor n/p \rfloor} \left[(-1)^{j+pk} \binom{n}{j+pk} - (-1)^{pk+(p-1)} \binom{n}{pk+(p-1)} \right]$$

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Looks complicated...

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j, p, n : fixed number

$k = 0$, compute $(*)$...

$k = 1$, compute $(*)$

$k = \lfloor n/p \rfloor$, compute $(*)$

Then, sum up everything.

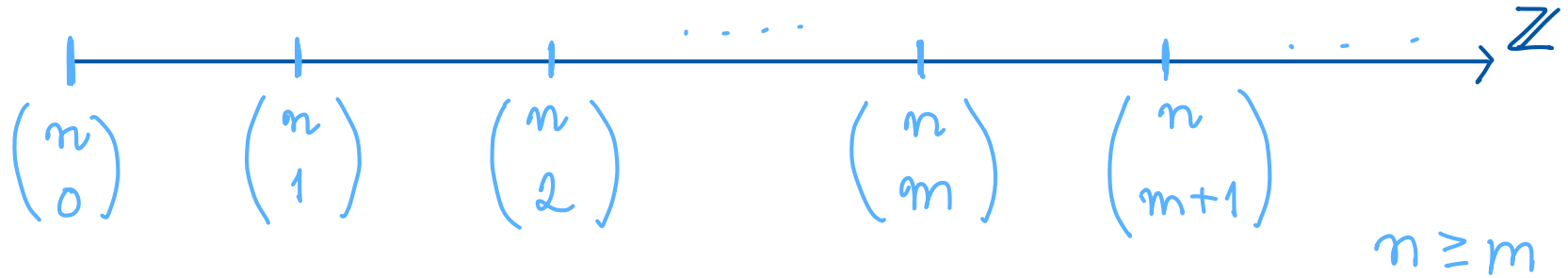
Binomial Expression - Cost

$$C_{j,n} = \sum_{k=0}^{\lfloor n/p \rfloor} \left[(-1)^{j+pk} \binom{n}{\underbrace{j+pk}_{O(n)}} - (-1)^{pk+(p-1)} \binom{n}{\underbrace{pk+(p-1)}_{O(n)}} \right]$$

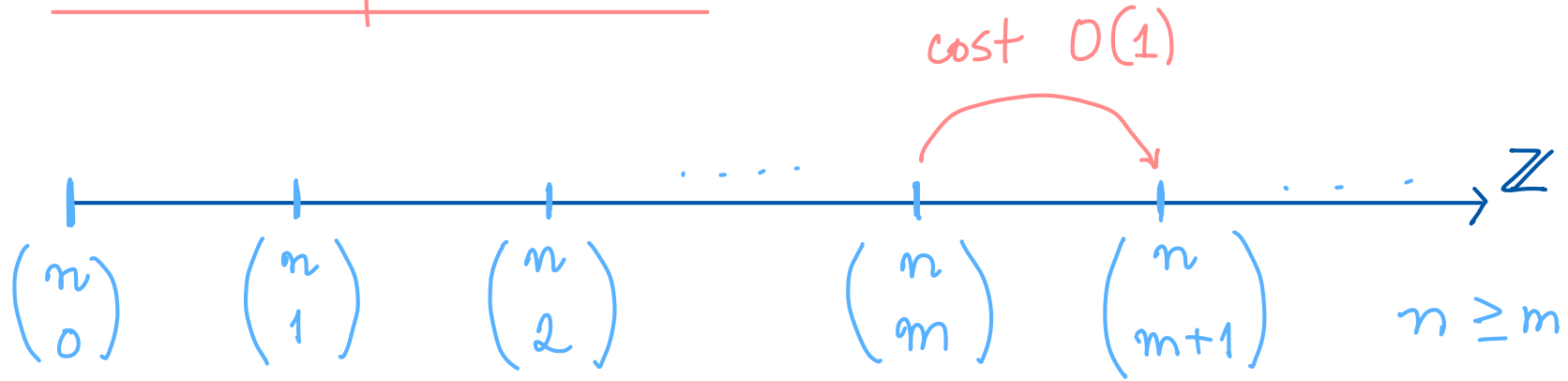
Perform $O(n)$ calculation $\lfloor n/p \rfloor$ times $\Rightarrow O(n^2)$

... but, can be smarter!

Binomial Expression - Cost



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Why?
$$\binom{n}{m+1} = \binom{n}{m} \frac{n-m}{m+1}.$$

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Prepare State: $\binom{n}{0} \xrightarrow{o(1)} \binom{n}{1} \xrightarrow{p(1)} \dots \rightarrow \binom{n}{j}$

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k=1.. and so on... In total, $\approx n$ steps!

Trigonometric Expansion

For $0 \leq j \leq p-2$:

$$c_{j,n} = \frac{1}{p} 2^{n+2} \left[\sum_{k=1}^{(p-1)/2} \sin\left(\frac{k\pi}{p}(p-1-j)\right) \sin^n\left(-\frac{k\pi}{p}\right) \cos\phi \right]$$

$$\phi = -\frac{2\pi k(p-1+j)+pk}{2p} + n \frac{\pi(2k+p)}{2p}.$$

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the summation is independent of n .
 p is a fixed number.

→ no matter how large " n " is, only needs $\frac{(p-1)}{2} - 1$ summation.

Trigonometric Expansion

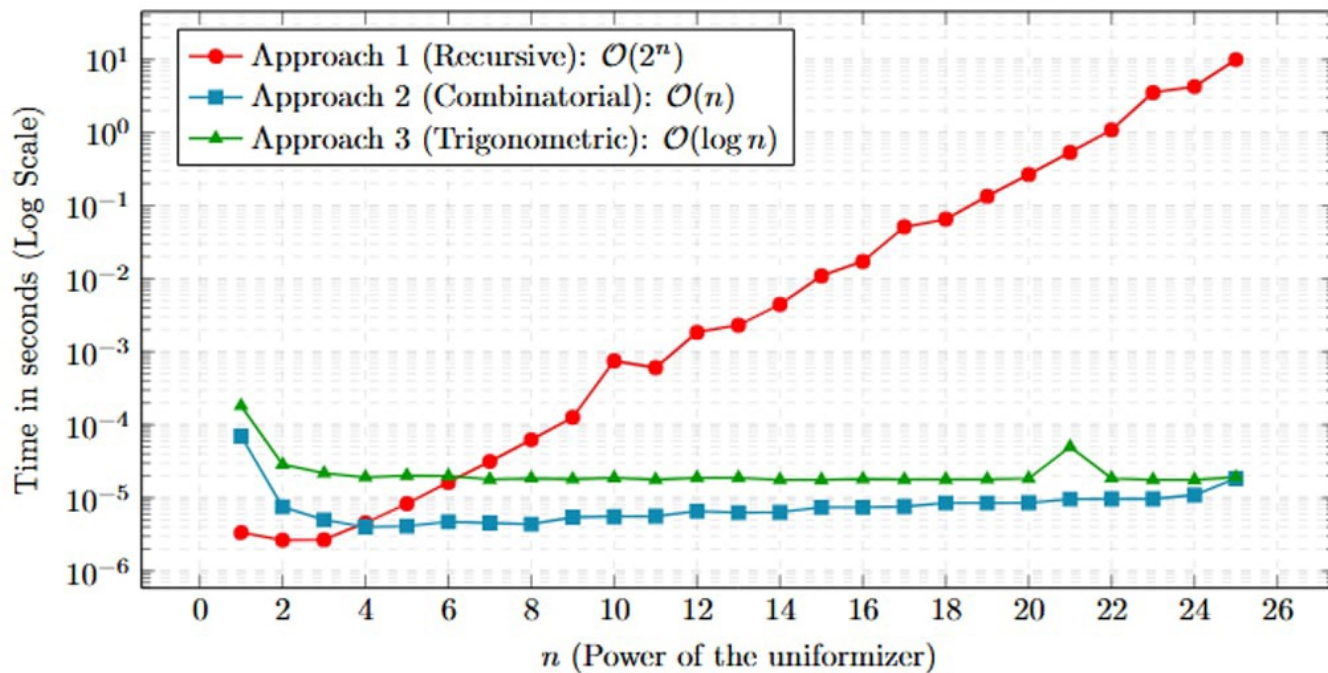
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Python: $\underbrace{O(\lg n)}$ or $\underbrace{O(1)}$ $\text{math.exp}(x)$.
** or np.pow()

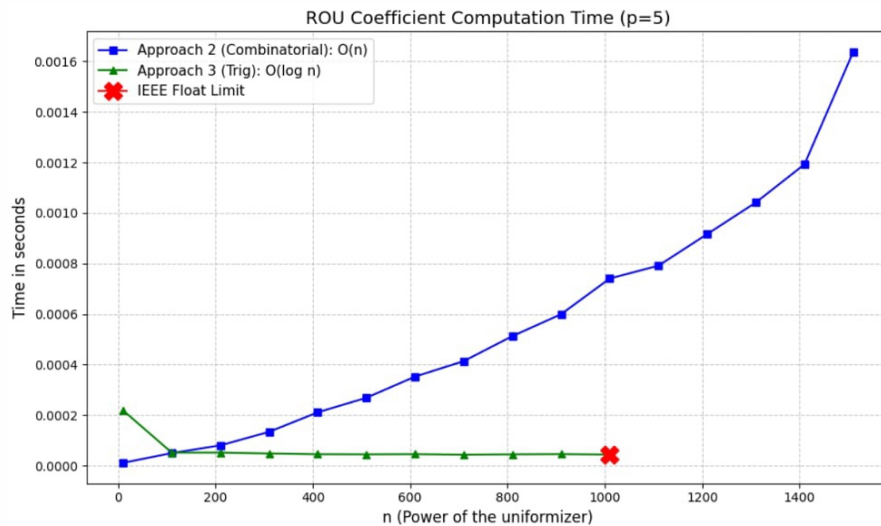
Numerical Result

ROU Coefficient Computation Time (Small n , $p = 3$)





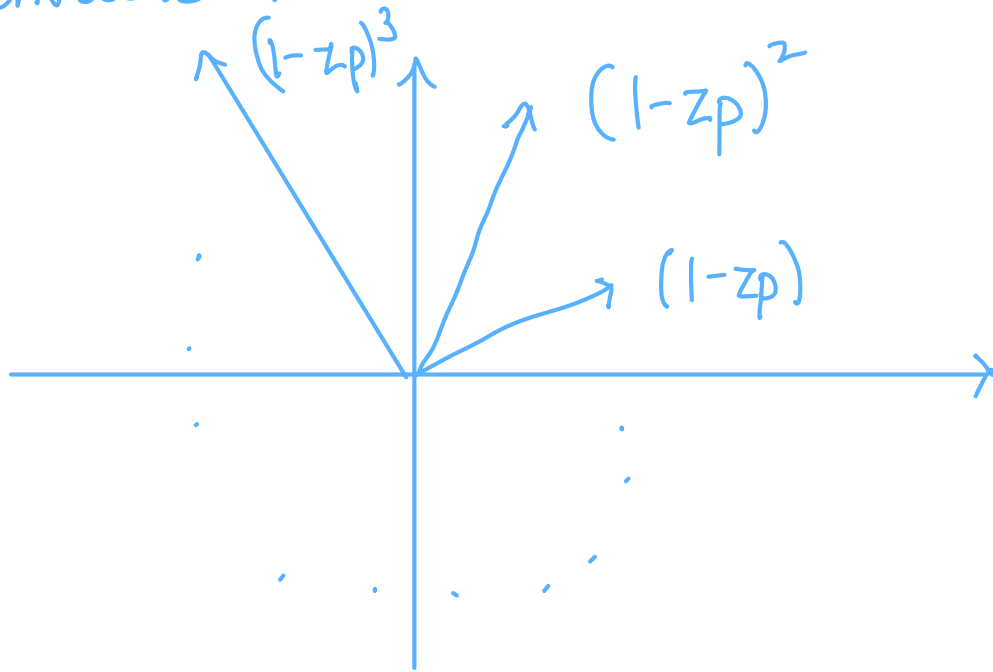
Hardware Limit



(CSC-330)

Future work

- Eigenvalue Problem



Future work

Conjecture: $(1-z_p)^n$ has a period of $p(p-1)$.

Back to the p -adic evaluation problem.

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(good luck contacting
me, I'm graduating)



GitHub

Citation



Randazzo, B. (2022).

Conductors for Maximal Orders of Group Rings over Local Fields
(Ph.D. dissertation).

Northern Illinois University.

[https://huskiecommons.lib.niu.edu/
allgraduate-thesesdissertations/7580/](https://huskiecommons.lib.niu.edu/allgraduate-thesesdissertations/7580/)



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